

Experiment 9 :

Implement a rule-based Part-of-Speech tagging system using regular expressions in Python.

Code:

```
import re

sentence = input("Enter a sentence: ")

words = sentence.split()

print("Rule-Based POS Tagging:")

for word in words:

    if re.fullmatch(r"*.ing", word):

        tag = "VBG"

    elif re.fullmatch(r"*.ed", word):

        tag = "VBD"

    elif re.fullmatch(r"*.ly", word):

        tag = "RB"

    elif word.lower() in ["the", "a", "an"]:

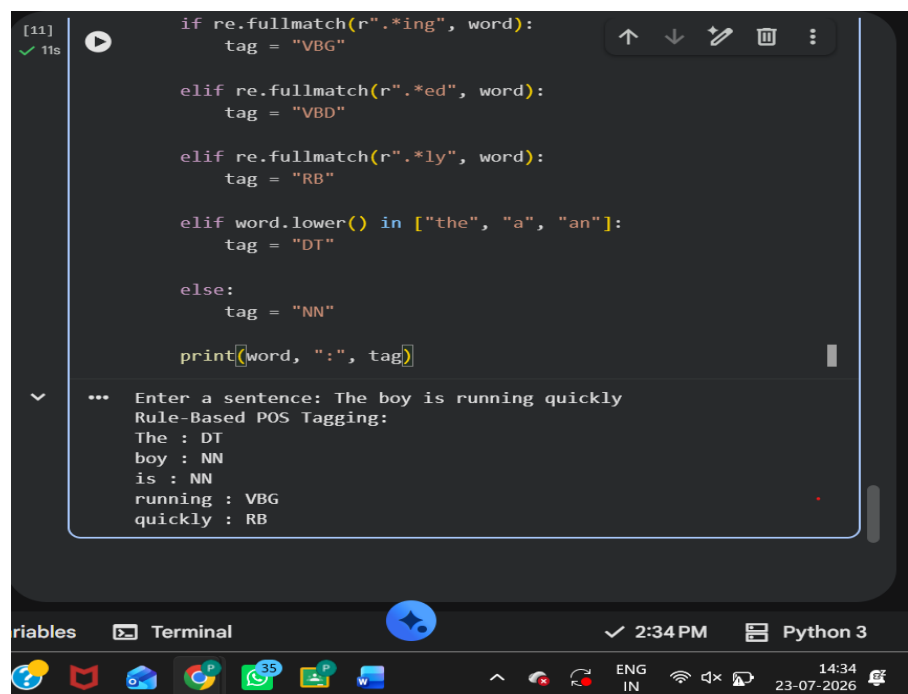
        tag = "DT"

    else:

        tag = "NN"

    print(word, ":", tag)
```

Output



```
[11] ✓ 11s
```

```
if re.fullmatch(r"*.ing", word):
    tag = "VBG"

elif re.fullmatch(r"*.ed", word):
    tag = "VBD"

elif re.fullmatch(r"*.ly", word):
    tag = "RB"

elif word.lower() in ["the", "a", "an"]:
    tag = "DT"

else:
    tag = "NN"

print(word, ":", tag)
```

```
... Enter a sentence: The boy is running quickly
Rule-Based POS Tagging:
The : DT
boy : NN
is : NN
running : VBG
quickly : RB
```

Variables Terminal 2:34 PM Python 3

14:34 23-07-2026